adapted on
Lynford L. Goddard
Brian T. Cunningham

Lectures 6-7

Boundary Conditions

Poisson's and Laplace's Equation

Victoria Shao
yangshao@Illinois.edu

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Laws of Electrostatics

$$\nabla \cdot \vec{\boldsymbol{D}} = \rho$$

$$\nabla \times \vec{\boldsymbol{E}} = 0$$

$$\vec{\boldsymbol{D}} = \epsilon_0 \vec{\boldsymbol{E}}$$

Or

$$\nabla \cdot \epsilon_0 \vec{\boldsymbol{E}} = \rho$$

$$\nabla \times \vec{\boldsymbol{E}} = 0$$

$\vec{\boldsymbol{D}}$ and $\vec{\boldsymbol{E}}$ will in general be continuous functions of position $\vec{\boldsymbol{r}}$ except at boundary surfaces (contain surface charge density $\rho_s(\vec{\boldsymbol{r}})$)

Maxwell's Eqns

Faraday's Law $\oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$ $\quad \nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$

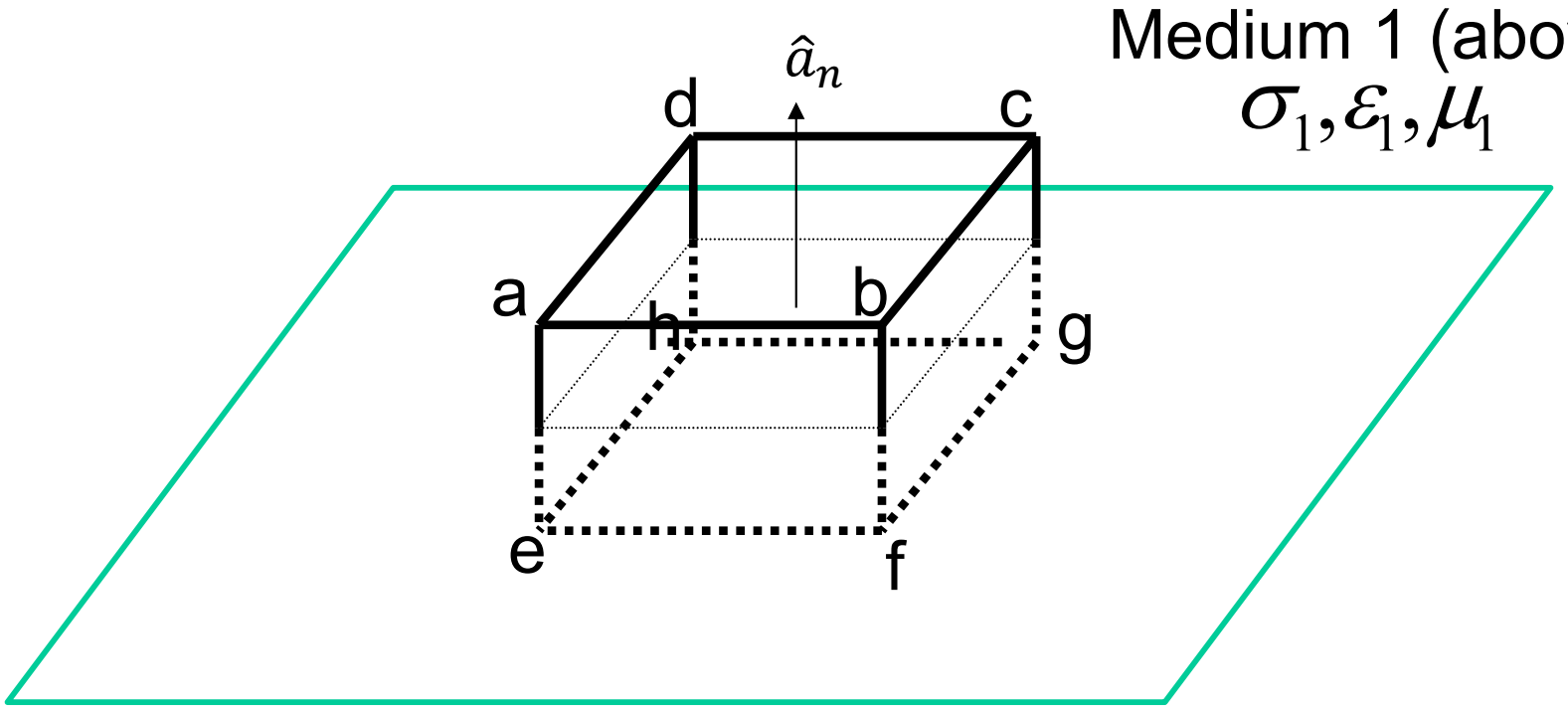
Ampere's Law $\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt}\iint_S \vec{D} \bullet d\vec{S}$ $\quad \nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$

Gauss' Law $\oiint_S \vec{B} \bullet d\vec{S} = 0$ $\quad \nabla \bullet \vec{B} = 0$

Gauss' Law $\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV$ $\quad \nabla \bullet \vec{D} = \rho$

Integral forms are valid for ALL closed paths and closed surfaces, EVEN WHEN THEY SPAN A BOUNDARY BETWEEN TWO MATERIALS

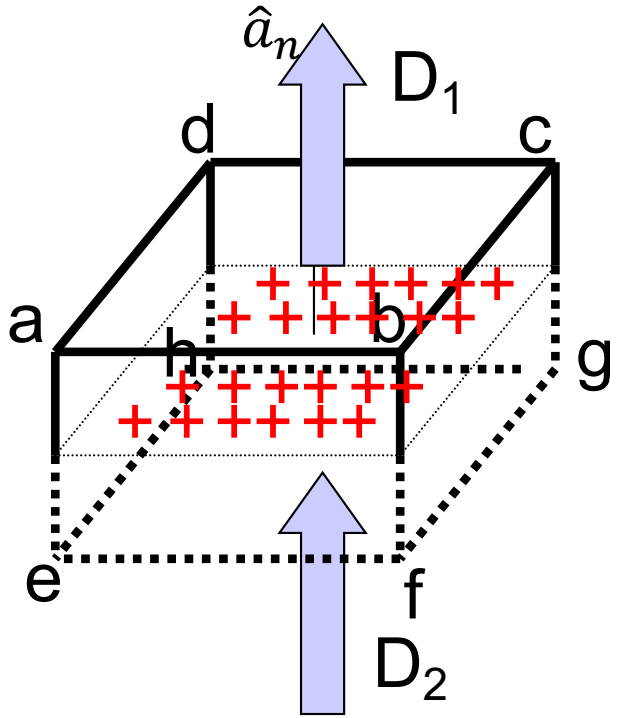
Closed surface through the volume



Closed surface: 6 sides of the cube

Apply Gauss's Law to the closed surface

Gauss' Law for D at Boundary



$$\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV$$

Take limit as *ae*, *bf*, *cg*, and *dh* go to zero

Consider D_1 and D_2 NORMAL TO SURFACE

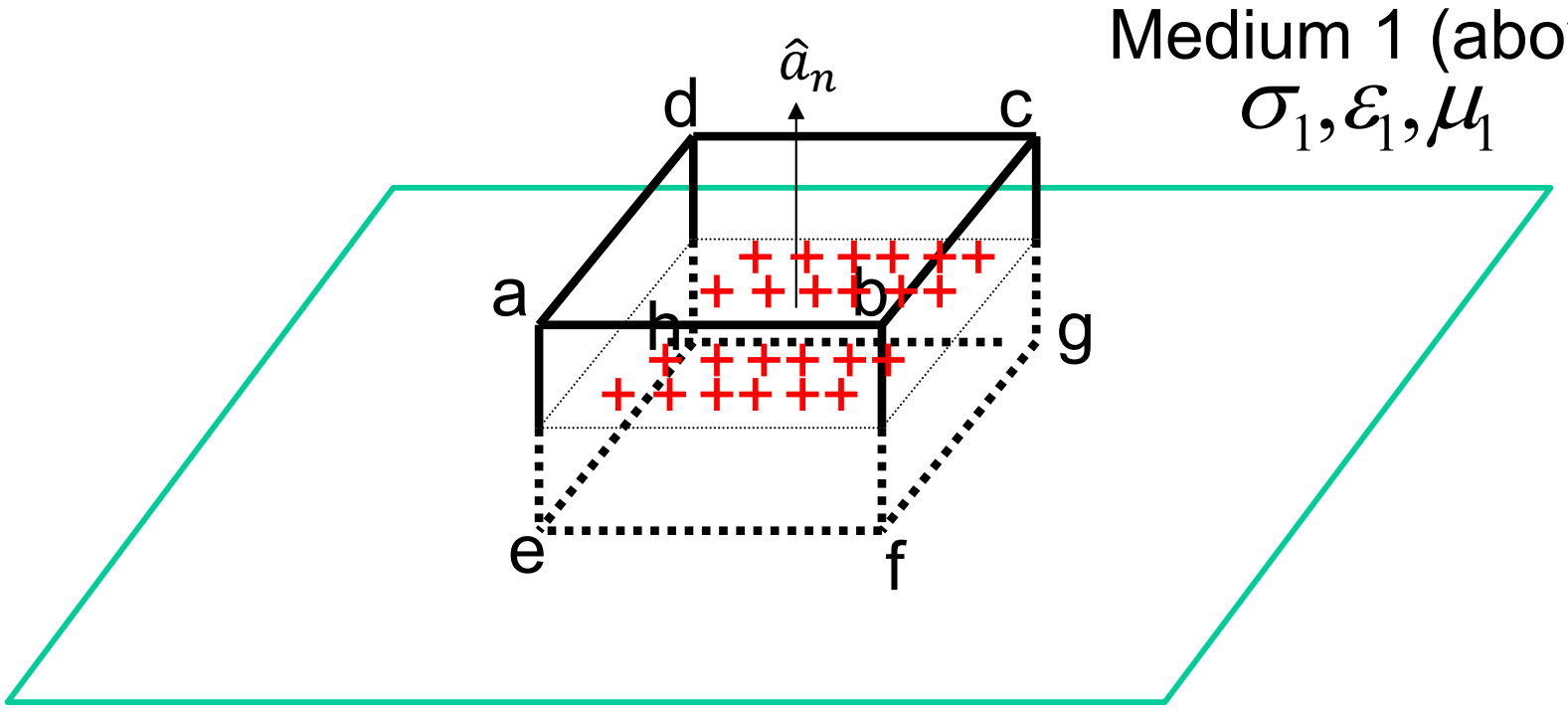
$$D_1 - D_2 = \rho \text{ i.e. } D_n \text{ is discontinuous because of } \rho_S$$

$$D_{n1} - D_{n2} = \rho_s$$

$$\hat{a}_n \cdot \left(\vec{D}_1 - \vec{D}_2\right) = \rho_s$$

Note: we know nothing about D_t

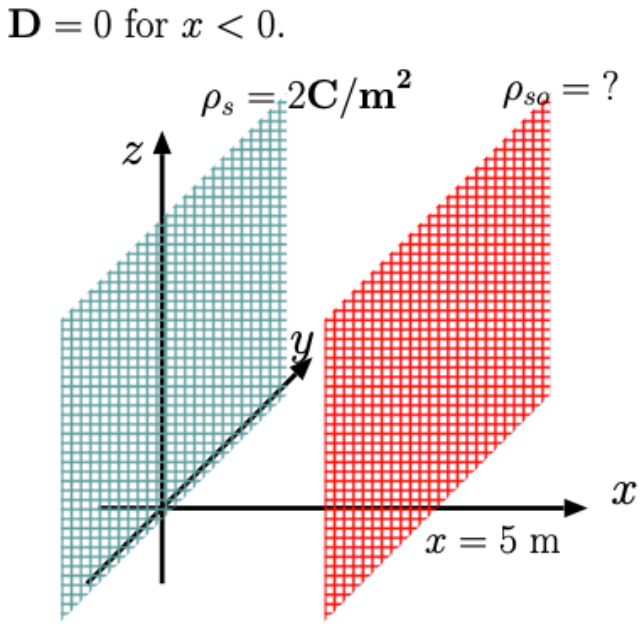
The closed volume can enclose surface charges



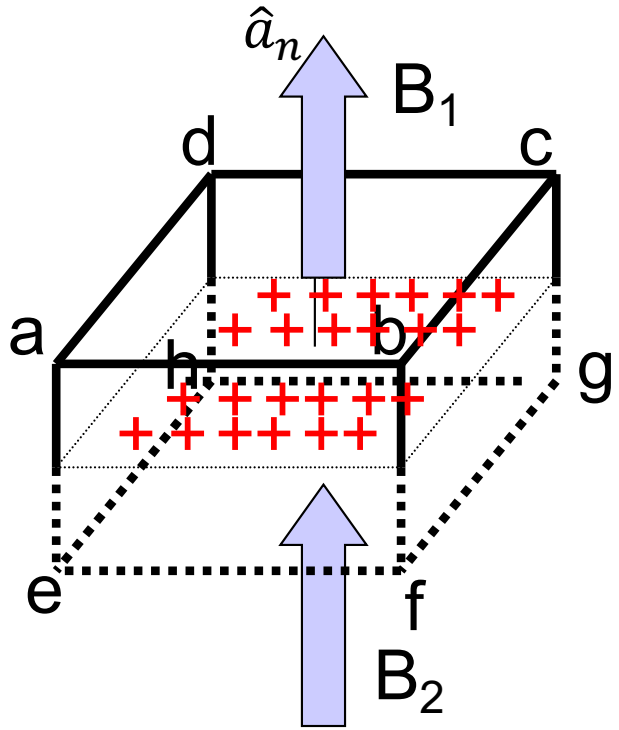
Example:
Free charges on the surface of a conductor

Finding D field

- Given $\vec{\boldsymbol{D}} = \mathbf{0}$ in the region $x < 0$ and $x > 5\,m$. Also, $x = 0$ and $x = 5$ m plane contain surface charge densities of $\rho_s = 2\,C/m^2$ and ρ_{so}. Determine ρ_{so} and $\vec{\boldsymbol{D}}$ for $0 < x < 5\,m$ if there are no other charge distributions.



Gauss' Law for B at Boundary



$$\oiint_S \vec{B} \bullet d\vec{S} = 0$$

Take limit as *ae*, *bf*, *cg*, and *dh* go to zero
Consider B_1 and B_2 NORMAL TO SURFACE

$$B_1 = B_2 \text{ i.e. } B_n \text{ is continuous}$$

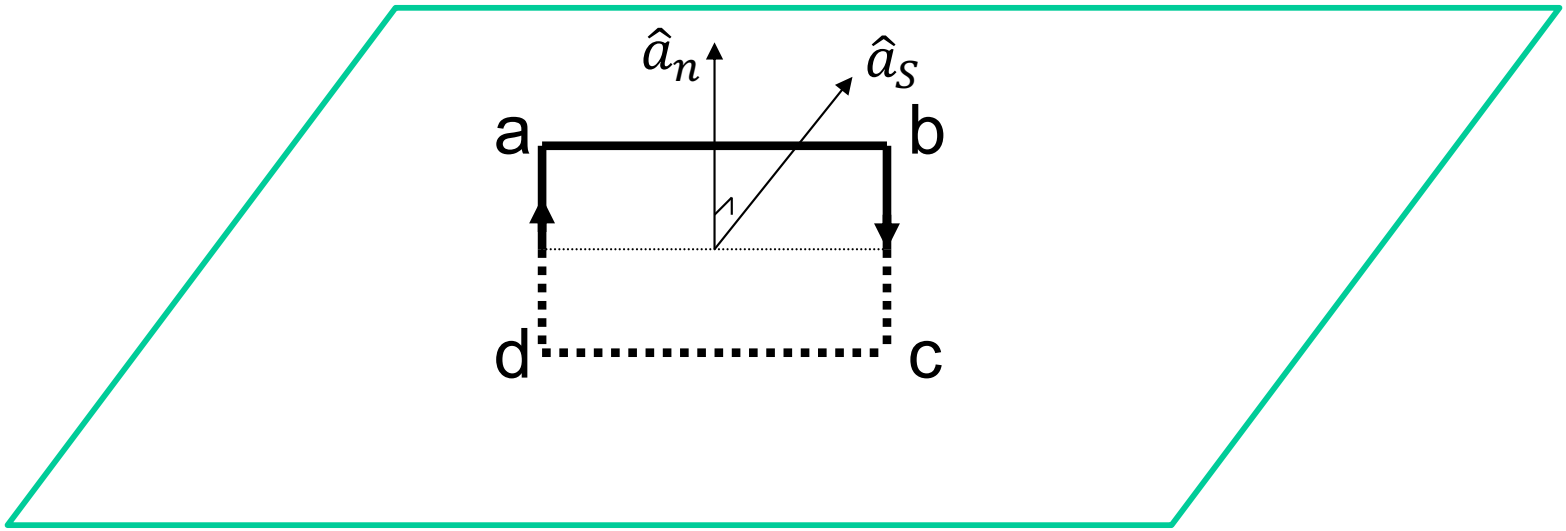
$$B_{n1} - B_{n2} = 0$$

$$\hat{a}_n \cdot \left(\vec{B}_1 - \vec{B}_2\right) = 0$$

Closed Path Through a Boundary

Medium 1 (above)

$\sigma_1, \varepsilon_1, \mu_1$



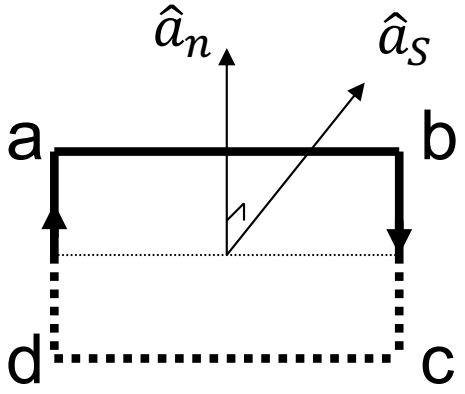
Medium 2 (below)

$\sigma_2, \varepsilon_2, \mu_2$

Closed path: abcd

Apply Faraday's Law and Ampere's Law to the closed path

Normal Vectors



Medium 1 (above)
$\sigma_1, \varepsilon_1, \mu_1$

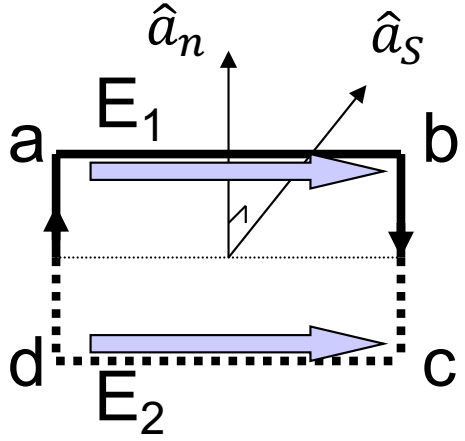
Medium 2 (below)
$\sigma_2, \varepsilon_2, \mu_2$

$\hat{\boldsymbol{a}}_n$ Vector NORMAL to the boundary. Points INTO medium 1

$\hat{\boldsymbol{a}}_S$ Vector normal to the path, TANGENT to the interface. Use right hand rule for path to define direction

Faraday's Law at Boundary

$$\oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S} = 0$$



Medium 1 (above)

$\sigma_1, \varepsilon_1, \mu_1$

Medium 2 (below)

$\sigma_2, \varepsilon_2, \mu_2$

Take limit as *ad* and *bc* go to zero

Consider remaining E_1 and E_2 TANGENT TO SURFACE

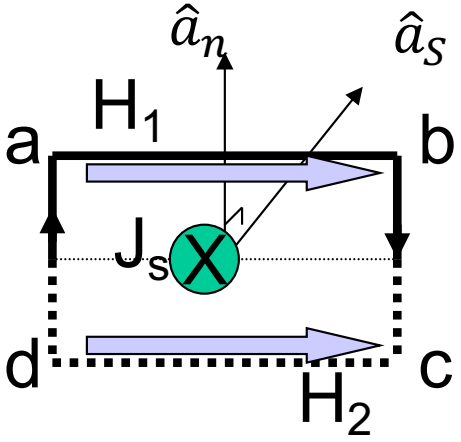
$$E_1 = E_2 \text{ i.e. } E_t \text{ is continuous}$$

$$E_{t1} - E_{t2} = 0$$

$$\hat{a}_n \times \left(\vec{E}_1 - \vec{E}_2\right) = 0$$

Ampere's Law at Boundary

$$\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S} = I_{enclosed}$$



Medium 1 (above)

$\sigma_1, \varepsilon_1, \mu_1$

Medium 2 (below)

$\sigma_2, \varepsilon_2, \mu_2$

Take limit as *ad* and *bc* go to zero

Consider remaining H_1 and H_2 TANGENT TO SURFACE

$$H_1 - H_2 = J_S \text{ i.e. } H_t \text{ is discontinuous because of } J_S$$

$$\vec{H}_{t1} - \vec{H}_{t2} = \vec{J}_s$$

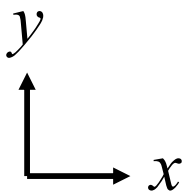
$$\hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$$

Note: we know nothing about H_n

Challenge Question: Boundary Conditions

- What is **$\mathbf{D_1}$**?

$$\varepsilon_1 = 1\varepsilon_0 \qquad \rho = 3\ C/m^2$$
$$\varepsilon_2 = 2\varepsilon_0 \qquad \vec{D}_2 = 3\hat{x} + 2\hat{y}\ C/m^2$$



$(a)\ \ 3\hat{x} + 2\hat{y}\ C/m^2$

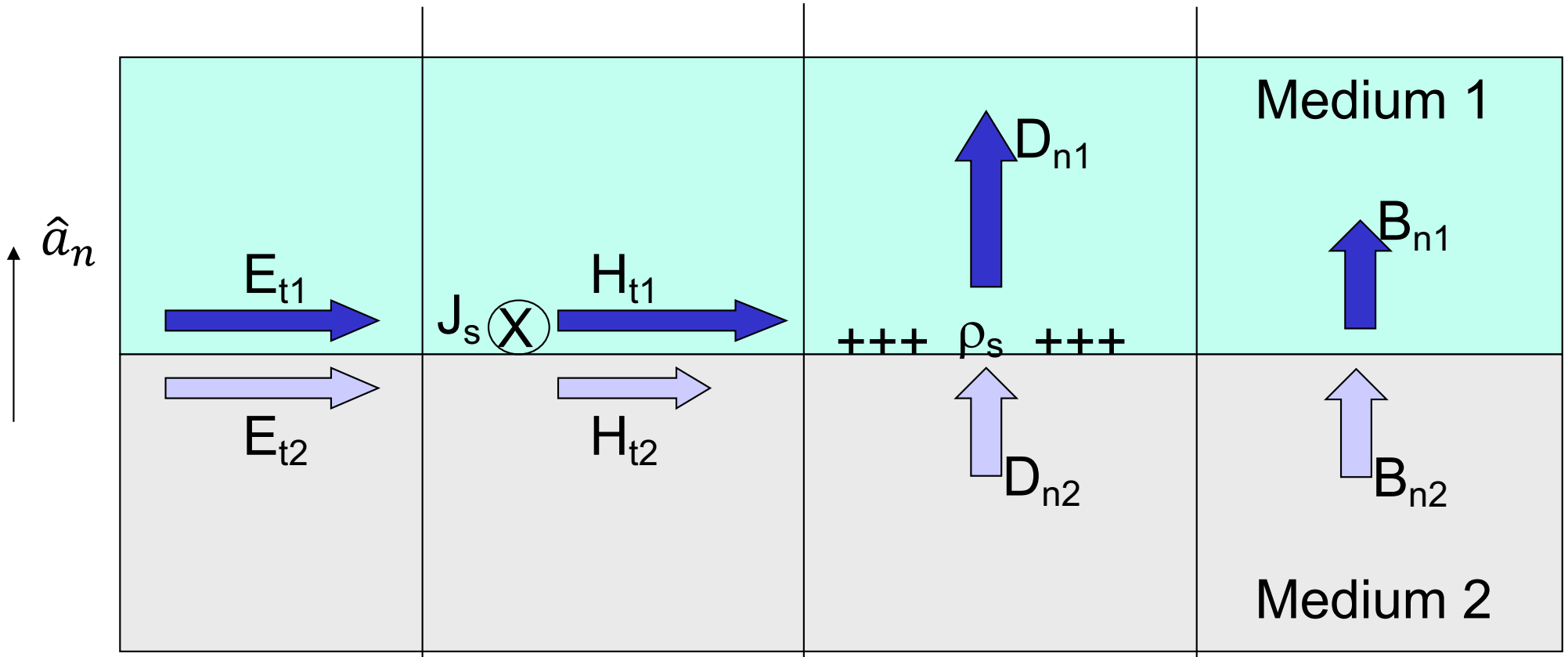
$(b)\ \ 3\hat{x} + 5\hat{y}\ C/m^2$

$(c)\ \ 3\hat{x} - 1\hat{y}\ C/m^2$

$(d)\ \ \frac{3}{2}\hat{x} + 5\hat{y}\ C/m^2$

$(e)\ \ \frac{3}{2}\hat{x} - 1\hat{y}\ C/m^2$

Remember this drawing!!



$$E_{t1} - E_{t2} = 0 \qquad \vec{H}_{t1} - \vec{H}_{t2} = \vec{J}_s \qquad D_{n1} - D_{n2} = \rho_s \qquad B_{n1} - B_{n2} = 0$$

$$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \qquad \hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \qquad \hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \qquad \hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$$

How to memorize BC?

Differential form of Maxwell's Equations		Boundary Conditions
$\nabla \cdot \vec{\boldsymbol{D}} = \rho$	⟶	$\hat{a}_n \cdot \left(\vec{D}_1 - \vec{D}_2\right) = \rho_s$
$\nabla \times \vec{\boldsymbol{E}} = -\dfrac{\partial \vec{\boldsymbol{B}}}{\partial t}$	⟶	$\hat{a}_n \times \left(\vec{E}_1 - \vec{E}_2\right) = 0$
$\nabla \cdot \vec{\boldsymbol{B}} = 0$	⟶	$\hat{a}_n \cdot \left(\vec{B}_1 - \vec{B}_2\right) = 0$
$\nabla \times \vec{\boldsymbol{H}} = \vec{\boldsymbol{J}} + \dfrac{\partial \vec{\boldsymbol{D}}}{\partial t}$	⟶	$\hat{a}_n \times \left(\vec{H}_1 - \vec{H}_2\right) = \vec{J}_s$

Laplacian Operator

$$\nabla \bullet \left(\nabla \Phi \right) = \nabla^2 \Phi$$

The "Laplacian" of a scalar field. (also called "Del Squared")

$\Phi(x,y,z)$ Scalar Field

$$\nabla \Phi = \frac{\partial \Phi}{\partial x} \hat{a}_x + \frac{\partial \Phi}{\partial y} \hat{a}_y + \frac{\partial \Phi}{\partial z} \hat{a}_z$$ Vector

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2}$$ Scalar

Poisson Equation for Potentials (static fields, constant ε)

$$\nabla \bullet \left(\nabla \Phi \right) = -\frac{\rho}{\varepsilon}$$

$$\nabla^2 \Phi = -\frac{\rho}{\varepsilon}$$

Known as "Poisson's Equation"

It's just Gauss' Law in terms of Potentials for a Static Field

If ρ=0, it is known as "Laplace's Equation"

Derivation of Poisson's & Laplace's equations

$$\nabla \bullet \mathbf{D} = \rho, \quad \mathbf{D} = \varepsilon \mathbf{E}$$

$$\mathbf{E} = -\nabla V$$

$$\nabla \bullet (\varepsilon \mathbf{E}) = -\nabla \bullet (\varepsilon \nabla V) = \rho$$

for homogeneous medium

$$\varepsilon \nabla \bullet \nabla V = \varepsilon \nabla^2 V = -\rho$$

$$\nabla \bullet \nabla V = \nabla \bullet \left(\hat{\mathbf{x}} \frac{\partial V}{\partial x} + \hat{\mathbf{y}} \frac{\partial V}{\partial y} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} \right)$$

$$= \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) \bullet \left(\hat{\mathbf{x}} \frac{\partial V}{\partial x} + \hat{\mathbf{y}} \frac{\partial V}{\partial y} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} \right)$$

$$= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\nabla^2 V = -\frac{\rho}{\varepsilon} \quad \text{(Poisson's equation)}$$

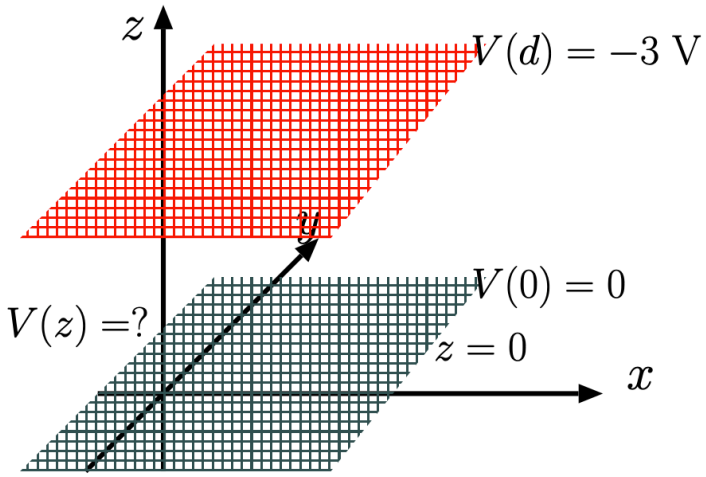
$$\nabla^2 V = 0 \quad \text{(Laplace's equation)}$$

Laplace operator has different forms for different coordinate systems.

Example

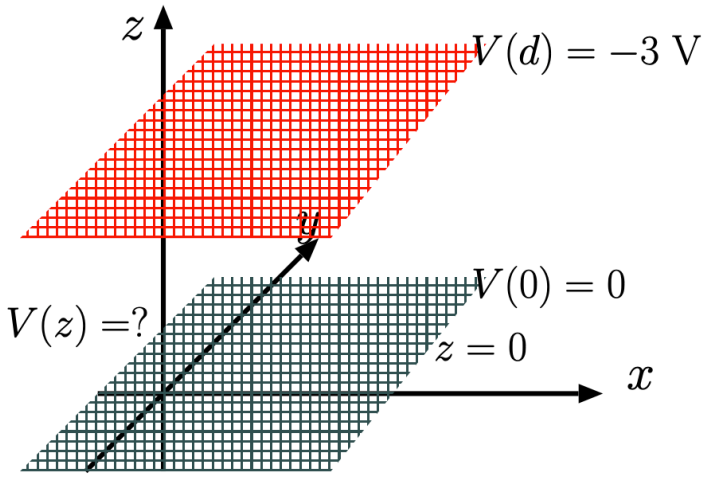
Consider a pair of parallel conducting metallic plates separated in z direction by distance of d = 2 m.

There is electrostatic field $\vec{\boldsymbol{E}} = E_z\hat{z}$ measured in between the plates. Using Laplace's equation determine V (z) and E(z) between the plates if the potential of the plate at z = 0 is 0 (the ground), while the potential of the plate at z = d is −3 V.

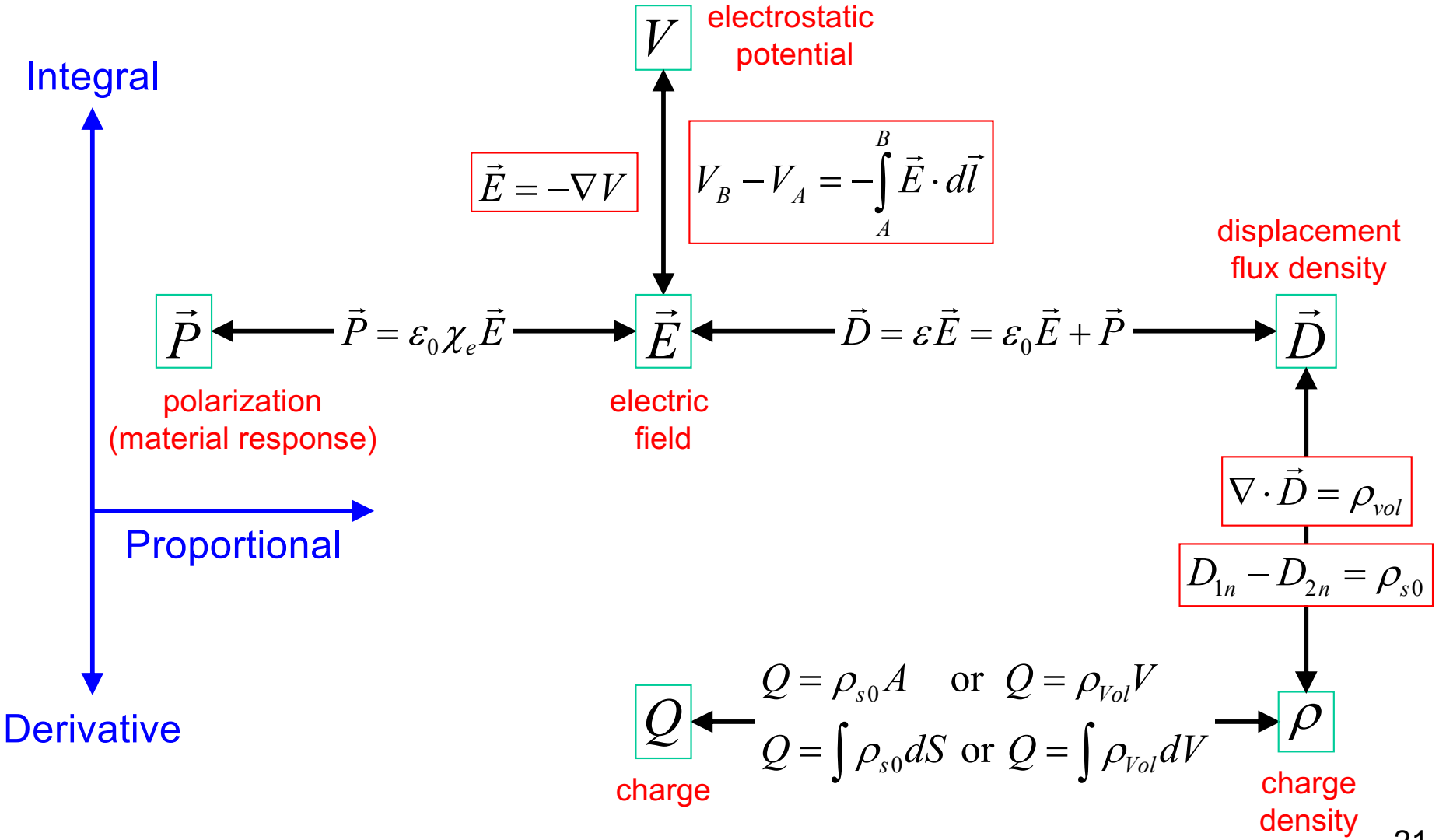


Example (cont.)

what are the surface charge densities of the metallic plates located at z = 0 and z = 2 m surfaces?



Connection of Concepts for Electrostatics



Summary

- Boundary conditions

$$E_{t1} - E_{t2} = 0 \qquad \vec{H}_{t1} - \vec{H}_{t2} = \vec{J}_s \qquad D_{n1} - D_{n2} = \rho_s \qquad B_{n1} - B_{n2} = 0$$

$$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \qquad \hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \qquad \hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \qquad \hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$$

- For static fields, if ε is constant, Φ satisfies Poisson's equation:

$$\nabla^2 \Phi = -\frac{\rho}{\varepsilon}$$